Build a Chatbot Using Hugging Face Transformers

```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM
model_name = "Qwen/Qwen2.5-1.5B-Instruct"
print("Loading model... ⏳ ")
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForCausalLM.from_pretrained(
    model_name,
    torch_dtype=torch.float32
)
model.to("cpu")
model.eval()
print("✅ Model loaded successfully!")
messages = [
    {
        "role": "system",
        "content": "You are a helpful AI assistant."
    }
]
def chatbot(user_input):
    global messages
    messages.append({"role": "user", "content": user_input})
    text = tokenizer.apply_chat_template(
        messages,
        tokenize=False,
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        add_generation_prompt=True
    )
    inputs = tokenizer(text, return_tensors="pt")
    with torch.no_grad():
        outputs = model.generate(
            **inputs,
            max_new_tokens=100,
            temperature=0.7,
            top_p=0.9,
            do_sample=True,
            pad_token_id=tokenizer.eos_token_id
        )
    response = tokenizer.decode(
        outputs[0][inputs["input_ids"].shape[-1]:],
        skip_special_tokens=True
    ).strip()
    messages.append({"role": "assistant", "content": response})
    return response
print("\n🤖 Chatbot Ready (CPU)")
print("Type 'exit' to quit | 'clear' to reset\n")
while True:
    user_input = input("You: ")
    if user_input.lower() in ["exit", "quit", "bye"]:
        print("Bot: Goodbye!")
        break
    if user_input.lower() == "clear":
```

```
        messages = [{"role": "system", "content": "You are a helpful AI assistant."}]
        print("Bot: Cleared.")
        continue
    print("Bot:", chatbot(user_input))
```